

Deciphering Wind Speed and Direction from Near-Sensor Antarctic Seismic Signals

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Abstract

Monitoring the weather in extreme and remote environments, such as Antarctica, remains a significant challenge. Conventional automatic weather stations are difficult to install and require frequent maintenance under harsh conditions. This study investigates a complementary approach to automatic weather stations that uses seismic signal processing to infer near-surface wind speed and direction.

Broadband seismometers, although designed to record distant seismic events, continuously capture high-frequency environmental noise generated by near-sensor processes. By inverting the conventional perspective and analysing this ambient high-frequency seismic noise, we identify signatures associated with local environmental forcing and examine correlations between seismic noise characteristics and wind speed and direction, using co-located automatic weather station data near Casey Station, East Antarctica.

Our results show that seismic noise contains measurable information about atmospheric dynamics, particularly during strong wind. Equipping seismic installations with minimal additional sensors, such as temperature and pressure gauges, enables them to serve as proxy weather stations. Given their robustness and suitability for long-term, large-scale deployment, seismic networks offer a promising complementary method for monitoring surface conditions in remote and extreme environments.

Introduction

Seismic signals are vibrations that propagate through the Earth (Kanamori and Brodsky, 2004). These vibrations can be recorded by seismometers, which are sensitive instruments capable of detecting small vibrations, such as earthquakes from the opposite side of the planet (Ringler and Bastien, 2020). Most of the signals recorded by seismometers are considered noise as they obstruct signals of interest (Bormann and Wielandt, 2013). Noise is typically filtered out in traditional seismology, however the analysis of noise can still provide a wealth of valuable information. Investigating near-sensor noise provides insights into the local environment, such as human activity, the surrounding ecosystem, and atmospheric processes. Modern frequency spectrum analysis has characterised seismic signals to be above the frequency of most inner-Earth processes, allowing us to isolate higher frequencies containing the near-sensor vibrations (Diaz, 2016). Wind-induced seismic noise has been investigated in regions outside of Antarctica by comparing the recorded seismic wind signals to data collected by an automatic weather station (AWS) (Lott et al., 2017; Dybing et al., 2019; Frankinet, Lecocq, and Camelbeeck, 2021). In this study, we examine correlations between seismic noise and AWS wind data and train random forest models to predict wind speed and direction based on seismic power measurements.

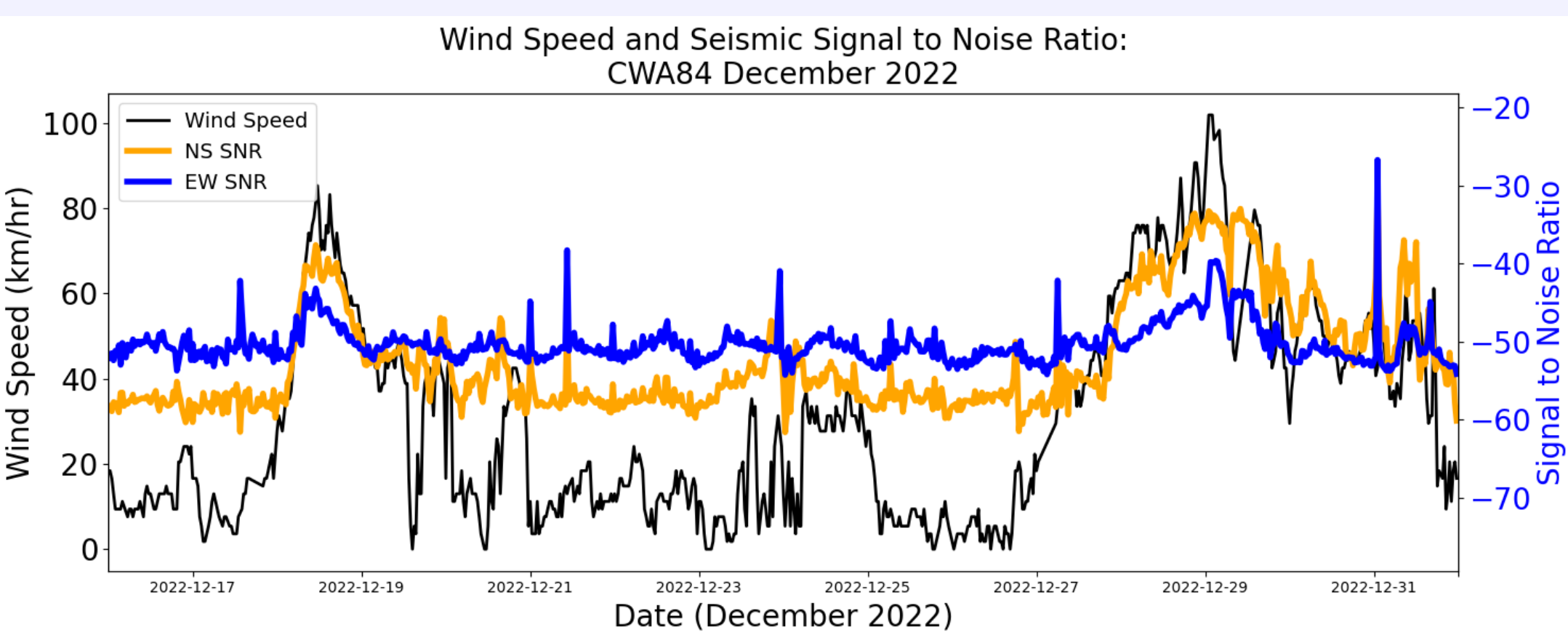


Figure 1. AWS wind speed measurements (black) displayed with the seismic signal (wind vibration) to noise ratio for the second half of December 2022, near Casey Station, Antarctica. More seismic signal occurs during windy periods. Orange: North-south seismic component; Blue: East-west seismic component.



Figure 2. Photo of the seismometer setup by Shyla Kupis. A solar panel sits on the snow surface providing power to the seismometer below.

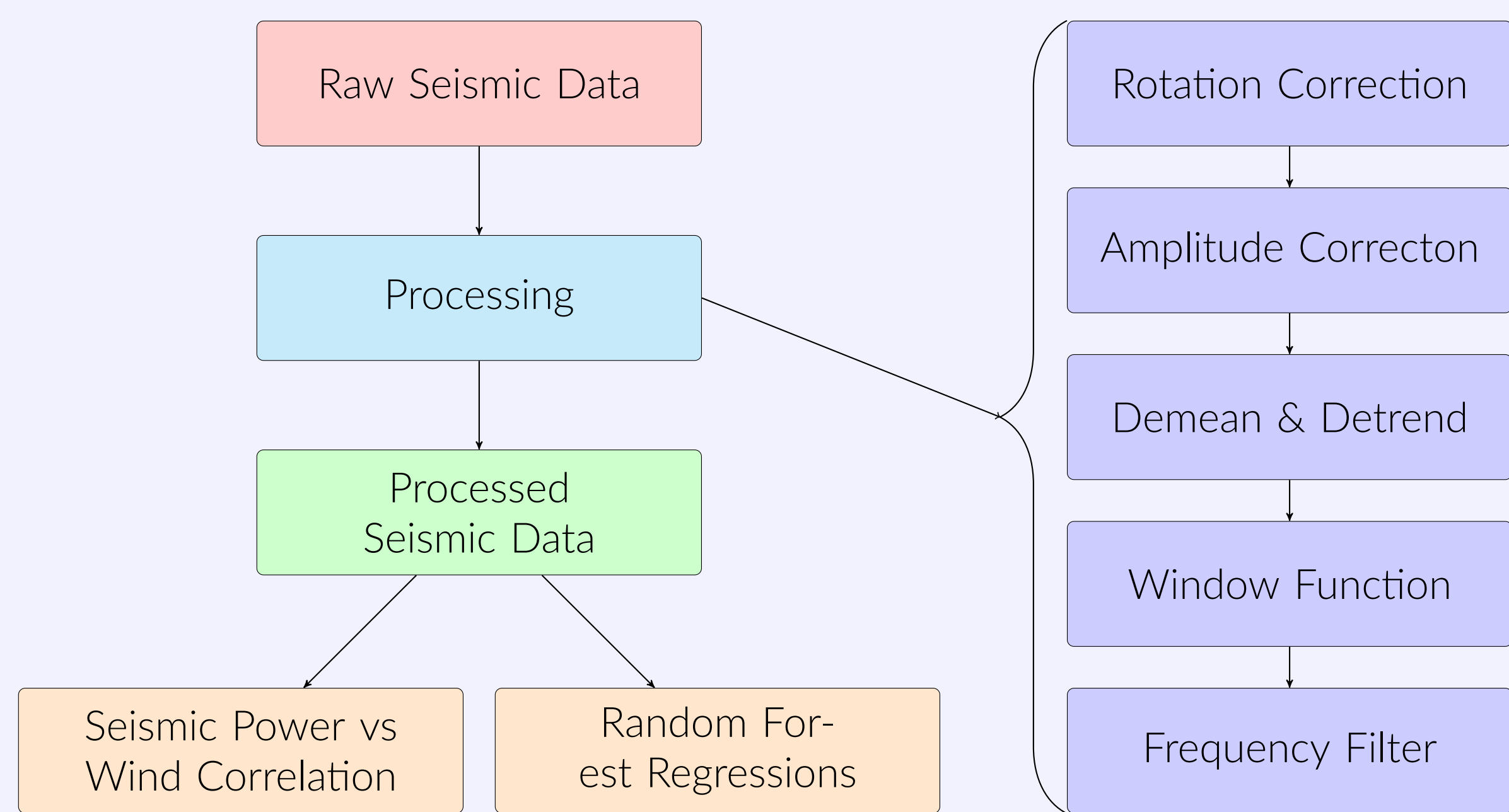


Figure 3. The seismometer type used for this study: Nanometrics Trillium Compact seismometer (Nanometrics Inc., 2026).

Research Aims

- Investigate the seismic power produced by wind.
- Build models to predict wind speed and direction based on seismic power.

Methodology



Code Statement

All of the code for this project can be found on the seismic-weather GitHub: <https://github.com/Ramirez873/seismic-weather>. This project and its codebase are continuously evolving as development progresses.



Seismic Power and Wind Speed

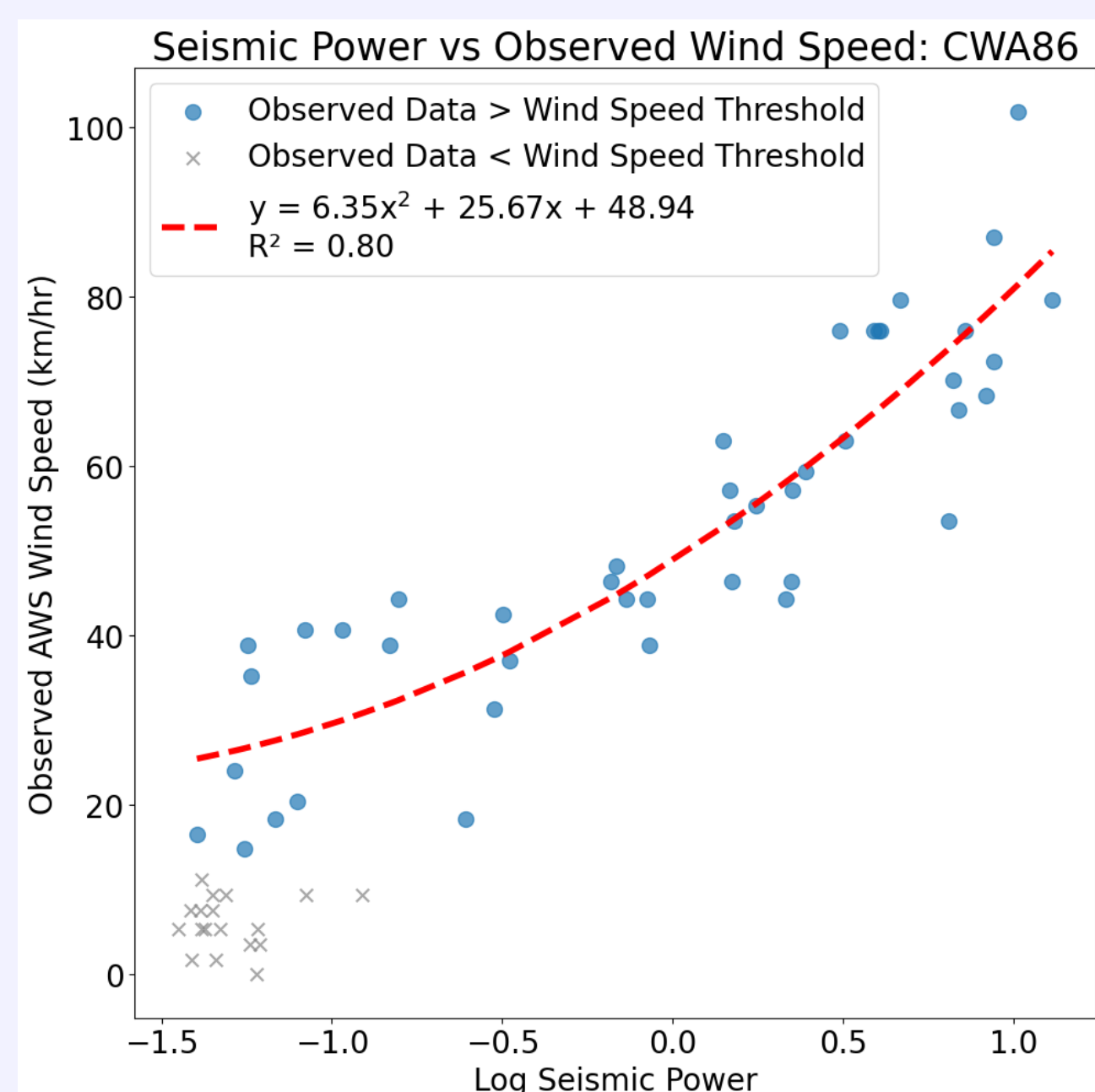


Figure 4. 13.5 Hz band NS seismic power from seismometer 'CWA86' vs nearby AWS wind speed measurements. Grey data points are from vibrations caused by wind speeds less than 15 km/hr and are not included in the regression line.

- Low wind speeds caused minimal vibration on the seismometer and produced results with high uncertainty.
- The relationship of the 13-14 Hz band log seismic power to observed AWS wind speed follows a positive quadratic relationship.
- The north-south component of the seismometer correlates with the wind speed more than the east-West

Random Forest Regression Results

- The 13-14 Hz frequency band was found to produce the optimal random forest permutation importance value.
- The random forest model generally predicts wind speed and direction similar to the measured observations.
- Low wind vibrational data (<15 km/hr, very low seismic power) lead to inaccurate predictions in wind speed and direction.

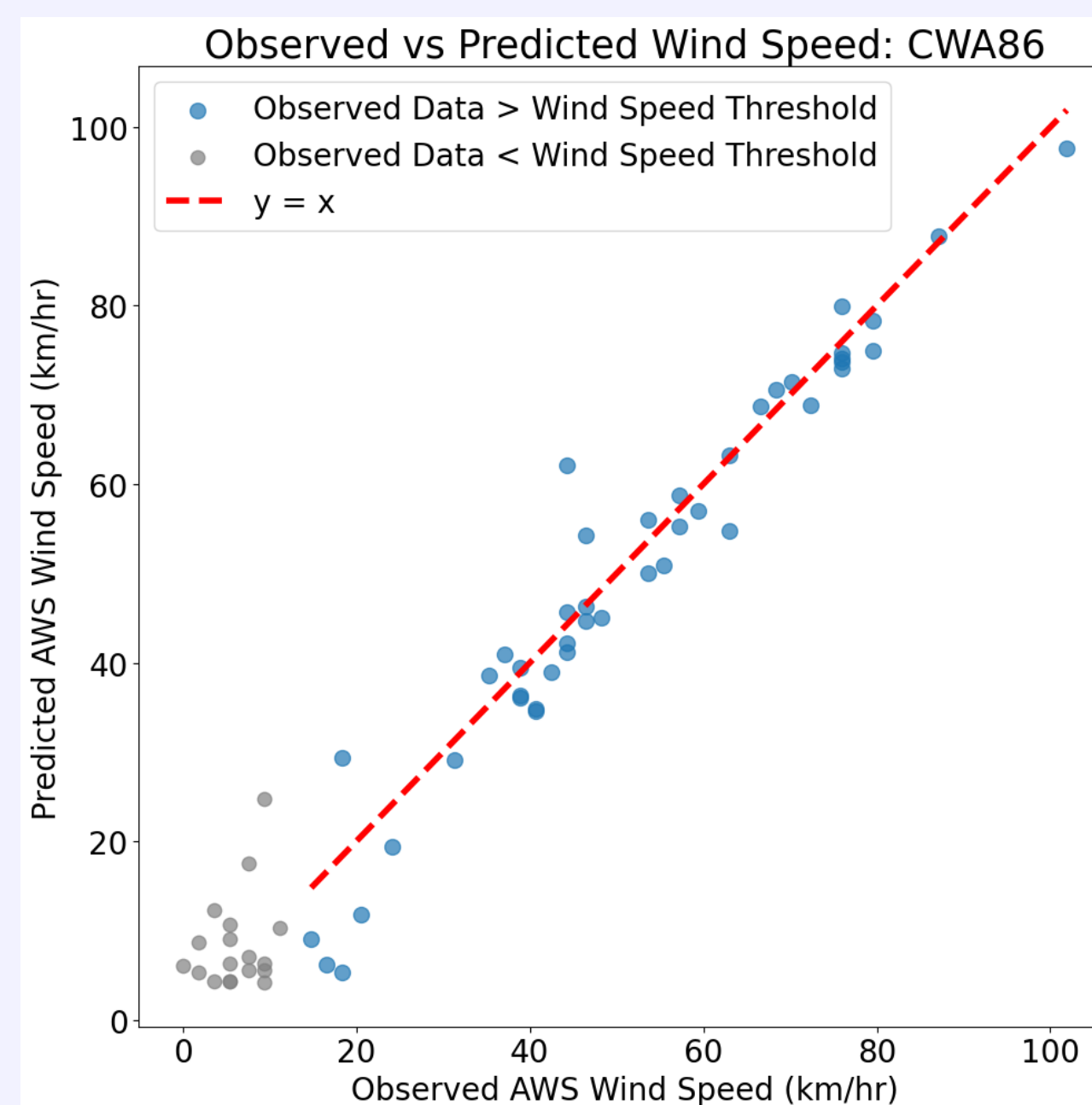


Figure 5. Observed AWS wind speed vs the random forest model predicted wind speed ($R^2 = 0.93$). Grey data points are from vibrations caused by wind speeds less than 15 km/hr and are not included in the line of best fit.

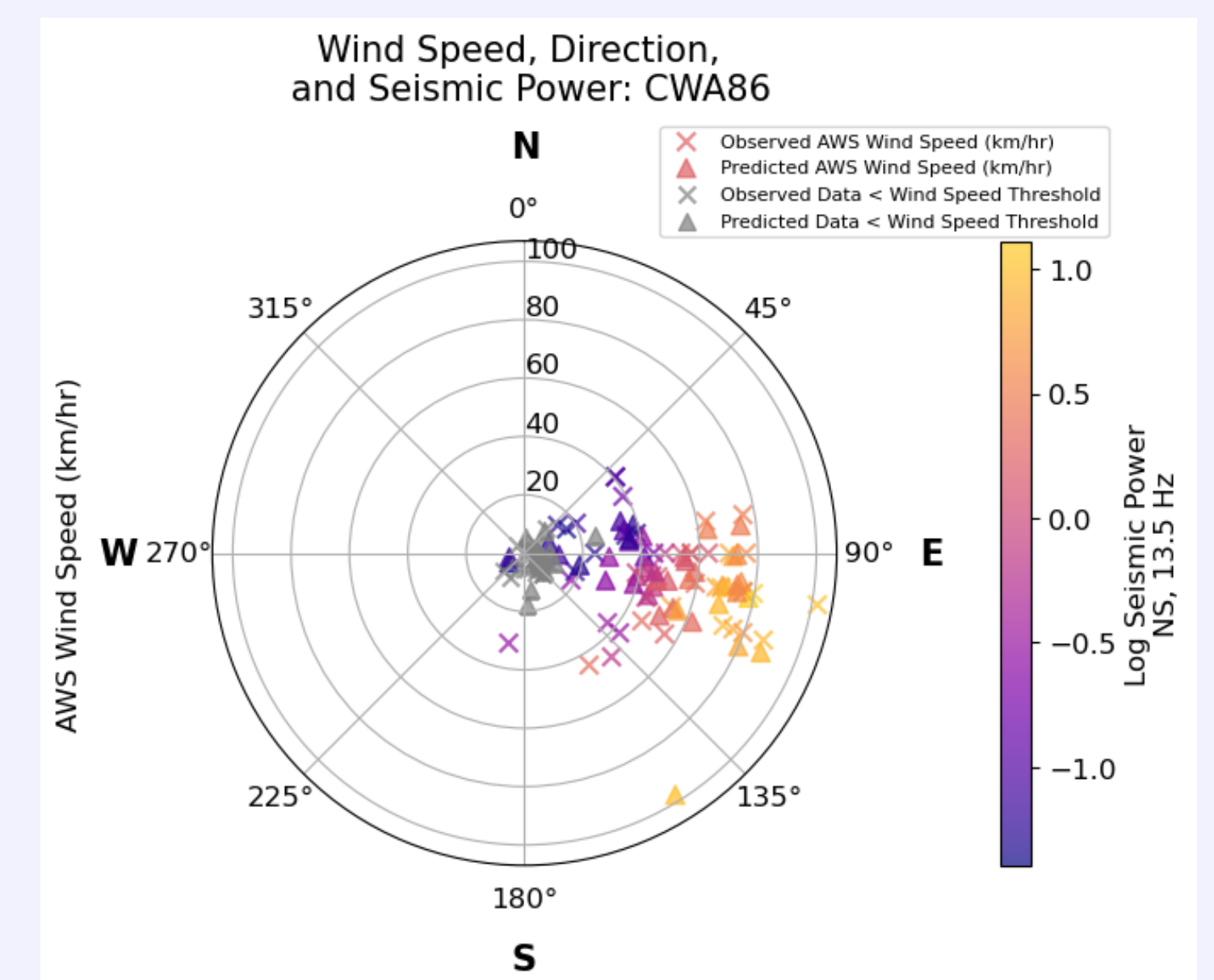


Figure 6. A polar plot showing the observed AWS wind measurements, 13.5 Hz band CWA86 seismic power, and the random forest model predicted measurements. Radial: Wind speed; Bearings: Wind direction; Colour: Seismic power; Markers: Observation or prediction. Grey data points are from vibrations caused by wind speeds less than 15 km/hr.

Discussion and Conclusion

- There is a high correlation between seismic power of specific frequencies and wind speed.
- Training a random forest model enables the wind speed and direction to be predicted accurately; at least for wind speeds above 15 km/hr.
- The relationship between wind speed and seismic signal varies with each station's configuration, including the placement of the seismic instrument, solar panels, and guy wires, as well as local surface conditions and topography. Ongoing work aims to develop station-specific regression models that account for these site-dependent influences.
- Currently, the 15km/hr wind speed threshold is a limitation of this workflow to use seismometers to measure wind. Future research focused on extracting wind data from low seismic powers is required to minimise this limitation.
- This study has shown the potential for seismometers to be used to observe wind of 15 km/hr and above.

References

- Bormann, Peter and Erhard Wielandt (2013). "Seismic Signals and Noise". en. In: *New Manual of Seismological Observatory Practice 2 (NMSOP2)*. Ed. by Peter Bormann. DOI: [10.2312/GFZ.NMSOP-2_CH4](https://doi.org/10.2312/GFZ.NMSOP-2_CH4). URL: https://gfzpublic.gfz.de/pubman/item/item_124248 (visited on 06/01/2026).
- Diaz, J. (Oct. 2016). "On the origin of the signals observed across the seismic spectrum". In: *Earth-Science Reviews* 161, pp. 224–232. ISSN: 0012-8252. DOI: [10.1016/j.earscirev.2016.07.006](https://doi.org/10.1016/j.earscirev.2016.07.006). URL: <https://www.sciencedirect.com/science/article/pii/S0012825216301684> (visited on 05/30/2026).
- Dybing, Sydney N. et al. (Mar. 2019). "Characteristics and Spatial Variability of Wind Noise on Near-Surface Broadband Seismometers". In: *Bulletin of the Seismological Society of America* 109.3, pp. 1082–1098. ISSN: 0037-1106. DOI: [10.1785/0120180227](https://doi.org/10.1785/0120180227). URL: <https://doi.org/10.1785/0120180227> (visited on 02/19/2026).
- Frankinet, Baptiste, Thomas Lecocq, and Thierry Camelbeeck (Oct. 2021). "Wind-induced seismic noise at the Princess Elisabeth Antarctica Station". English. In: *The Cryosphere* 15.10, pp. 5007–5016. ISSN: 1994-0416. DOI: [10.5194/tc-15-5007-2021](https://doi.org/10.5194/tc-15-5007-2021). URL: <https://tc.copernicus.org/articles/15/5007/2021/> (visited on 03/04/2026).
- Kanamori, Hiroo and Emily E. Brodsky (July 2004). "The physics of earthquakes". en. In: *Reports on Progress in Physics* 67.8, p. 1429. ISSN: 0034-4885. DOI: [10.1088/0034-4885/67/8/R03](https://doi.org/10.1088/0034-4885/67/8/R03) (visited on 06/01/2026).
- Lott, Friederike F. et al. (Mar. 2017). "On the Analysis of Wind-Induced Noise in Seismological Recordings". en. In: *Pure and Applied Geophysics* 174.3, pp. 1453–1470. ISSN: 1420-9136. DOI: [10.1007/s00024-017-1477-2](https://doi.org/10.1007/s00024-017-1477-2). URL: <https://doi.org/10.1007/s00024-017-1477-2> (visited on 02/19/2026).
- Nanometrics Inc. (2026). *Trillium Compact* | Nanometrics. en-ca. URL: <https://nanometrics.ca/instrumentation/products/seismometers/trillium-compact> (visited on 06/03/2026).
- Ringler, Adam T. and Patrick Bastien (Jan. 2020). "A Brief Introduction to Seismic Instrumentation: Where Does My Data Come From?" In: *Seismological Research Letters* 91.2A, pp. 1074–1083. ISSN: 0895-0695. DOI: [10.1785/0220190214](https://doi.org/10.1785/0220190214). URL: <https://doi.org/10.1785/0220190214> (visited on 06/01/2026).